

CinemaPlus Database Structure

Tables, key columns, constraints, and current relationships.

Users		
Stores registered system users, including admins and clients.		
Column	Type	Rules
Id	int	PK, identity, required
FullName	varchar(150)	required
Email	varchar(180)	required, unique
Phone	varchar(30)	nullable
ProfilePhotoPath	varchar(300)	nullable
PasswordHash	varchar(255)	required
Role	varchar(30)	required
CreatedAt	datetime	required
Constraints / notes		
- Unique index: Email		
- Referenced by Bookings.UserId		

Films		
Movie catalogue entries available now or coming soon.		
Column	Type	Rules
Id	int	PK, identity, required
Title	varchar(200)	required, unique
Genre	varchar(100)	required
DurationMinutes	int	required
Description	text	required
ReleaseYear	int	required
AgeRestriction	varchar(20)	required
PosterPath	varchar(300)	required
TrailerPath	varchar(300)	nullable
AvailabilityDate	date	required
AvailabilityStatus	varchar(40)	required
Constraints / notes		
- Unique index: Title		
- Composite index: AvailabilityStatus + AvailabilityDate		
- Referenced by Sessions.FilmId		

Halls

Cinema halls with technology and seat-layout parameters.

Column	Type	Rules
Id	int	PK, identity, required
Name	varchar(80)	required, unique
Technology	varchar(80)	required
RowCount	int	required
SeatsPerRow	int	required

Constraints / notes

- Unique index: Name
- Referenced by Seats.HallId and Sessions.HallId

Seats

Physical seat map for every hall.

Column	Type	Rules
Id	int	PK, identity, required
HallId	int	FK -> Halls.Id, required
RowNumber	int	required
SeatNumber	int	required

Constraints / notes

- Unique index: HallId + RowNumber + SeatNumber
- Cascade delete from Halls
- Referenced by BookedSeats.SeatId

Sessions

Concrete showtimes for films in halls.

Column	Type	Rules
Id	int	PK, identity, required
FilmId	int	FK -> Films.Id, required
HallId	int	FK -> Halls.Id, required
SessionTime	datetime	required
Price	decimal(10,2)	required

Constraints / notes

- Unique index: FilmId + HallId + SessionTime
- Index: SessionTime
- Delete restricted from Films and Halls
- Referenced by Bookings.SessionId and BookedSeats.SessionId

Bookings

Customer orders for one session, optionally linked to a user account.

Column	Type	Rules
Id	int	PK, identity, required
UserId	int	FK -> Users.Id, nullable
SessionId	int	FK -> Sessions.Id, required
BookingDate	datetime	required
Status	varchar(40)	required
TotalPrice	decimal(10,2)	required
TicketCode	varchar(80)	required, unique
CustomerName	varchar(150)	required
CustomerEmail	varchar(180)	required
CustomerPhone	varchar(30)	nullable

Constraints / notes

- Unique index: TicketCode
- Index: BookingDate
- User delete behavior: SET NULL
- Session delete behavior: RESTRICT
- Referenced by BookedSeats.BookingId and NotificationLogs.BookingId

BookedSeats

Bridge table that reserves specific seats for a booking within a session.

Column	Type	Rules
Id	int	PK, identity, required
BookingId	int	FK -> Bookings.Id, required
SessionId	int	FK -> Sessions.Id, required
SeatId	int	FK -> Seats.Id, required

Constraints / notes

- Unique index: SessionId + SeatId
- Unique index: BookingId + SeatId
- Booking delete behavior: CASCADE
- Session and Seat delete behavior: RESTRICT

NotificationLogs

Audit trail of booking-related notifications and emails.

Column	Type	Rules
Id	int	PK, identity, required
BookingId	int	FK -> Bookings.Id, required
Email	varchar(180)	required
Message	text	required
CreatedDate	datetime	required
Status	varchar(40)	required

Constraints / notes

- Booking delete behavior: CASCADE

Table Relations

Logical relations currently implemented in the EF Core model.

Current Entity Relationships			
Parent	Child	Cardinality	Meaning
Users	Bookings	1 -> many	Bookings.UserId is nullable for guest orders; on user delete value becomes null
Films	Sessions	1 -> many	Deleting a film is restricted while sessions exist.
Halls	Seats	1 -> many	Deleting a hall cascades to its seat map.
Halls	Sessions	1 -> many	Deleting a hall is restricted while sessions exist.
Sessions	Bookings	1 -> many	Each booking belongs to exactly one session.
Bookings	BookedSeats	1 -> many	A booking may contain multiple seats.
Sessions	BookedSeats	1 -> many	Used to enforce unique sold seat per session.
Seats	BookedSeats	1 -> many	Physical seats can appear in many historical bookings across sessions.
Bookings	NotificationLogs	1 -> many	Notification history is removed when a booking is deleted.